

SRE STUDIUM PRZYPADKU

AZURE ARC HYBRID CLOUD LAB

Czesci 1-10: K8s, Azure Arc, Terraform, CI/CD, Ingress TLS, HPA, NetworkPolicy, RBAC, GitOps, Trivy, Defender, cert-manager, KEDA, Helm, OPA/Gatekeeper, Chaos Mesh, Linkerd

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IT Solutions Bartosz Suszko, Olsztyn

Data: Czerwiec 2026 (Czesci 1-10)

GitHub: github.com/buth11/sre-homelab-azure-arc

Spis treści

- 1–11. Kontekst, Infrastruktura, FinOps, Azure Arc, Observability, KQL, LB, Prometheus, Terraform, Policy, GitHub
- 12. Czesc 2: Service Principal, Remote State, CI/CD
- 13. Czesc 3: Ingress TLS, HPA, NetworkPolicy, RBAC
- 14. Postmortem POST-001
- 15. Wyniki i wnioski (Czesci 1-3)
- 16. Troubleshooting Log (10 incydentow)

- 17. Czesc 4: GitOps — FluxCD 2.8.8**
- 18. Czesc 4: Trivy — skanowanie CVE**
- 19. Czesc 4: Defender for Cloud**
- 20. Czesc 5: cert-manager — automatyczny TLS**
- 21. Czesc 6: KEDA — Event-Driven Autoscaling**
- 22. Czesc 7: Helm — własny chart aras-demo-app**
- 23. Czesc 8: OPA/Gatekeeper — audit polityk**
- 24. Czesc 9: Chaos Engineering — Chaos Mesh**
- 25. Czesc 10: Linkerd — Service Mesh, mTLS**
- 26. Wyniki i wnioski (Czesci 1-10)**
- 27. Troubleshooting Log (17 incydentow)**

Rozdział 1: Kontekst i architektura

Projekt demonstruje budowę produkcyjnego środowiska hybrydowego na GMKtec EVO-X2 (128 GB RAM) łączącego Kubernetes z Microsoft Azure przez Azure Arc. Trzy tygodnie praktycznych ćwiczeń z dokumentacją każdego kroku i 10 udokumentowanych incydentów.

Warstwa	Technologia	Zastosowanie
Kubernetes	K3s v1.35.4+k3s1	Klaster 3 węzły
Ingress	Traefik + TLS	HTTPS routing po domenie
Autoscaling	HPA (autoscaling/v2)	Skalowanie CPU 2-8 replik
Network Security	NetworkPolicy	Firewall wewnątrz klastra
Access Control	RBAC + ServiceAccount	Least privilege
Cloud Bridge	Azure Arc	Hybrydowa płaszczyzna kontrolna
Monitoring	Azure Monitor + Prometheus	Observability
IaC	Terraform + Remote State	Infrastructure as Code
CI/CD	Azure DevOps	Pipeline z approval gate

Rozdział 2: Infrastruktura On-Premise

Wezel	IP	CPU	RAM
k3s-master (Control Plane)	192.168.122.10	4 vCPU	8 GB
k3s-worker1	192.168.122.11	4 vCPU	16 GB
k3s-worker2	192.168.122.12	4 vCPU	16 GB

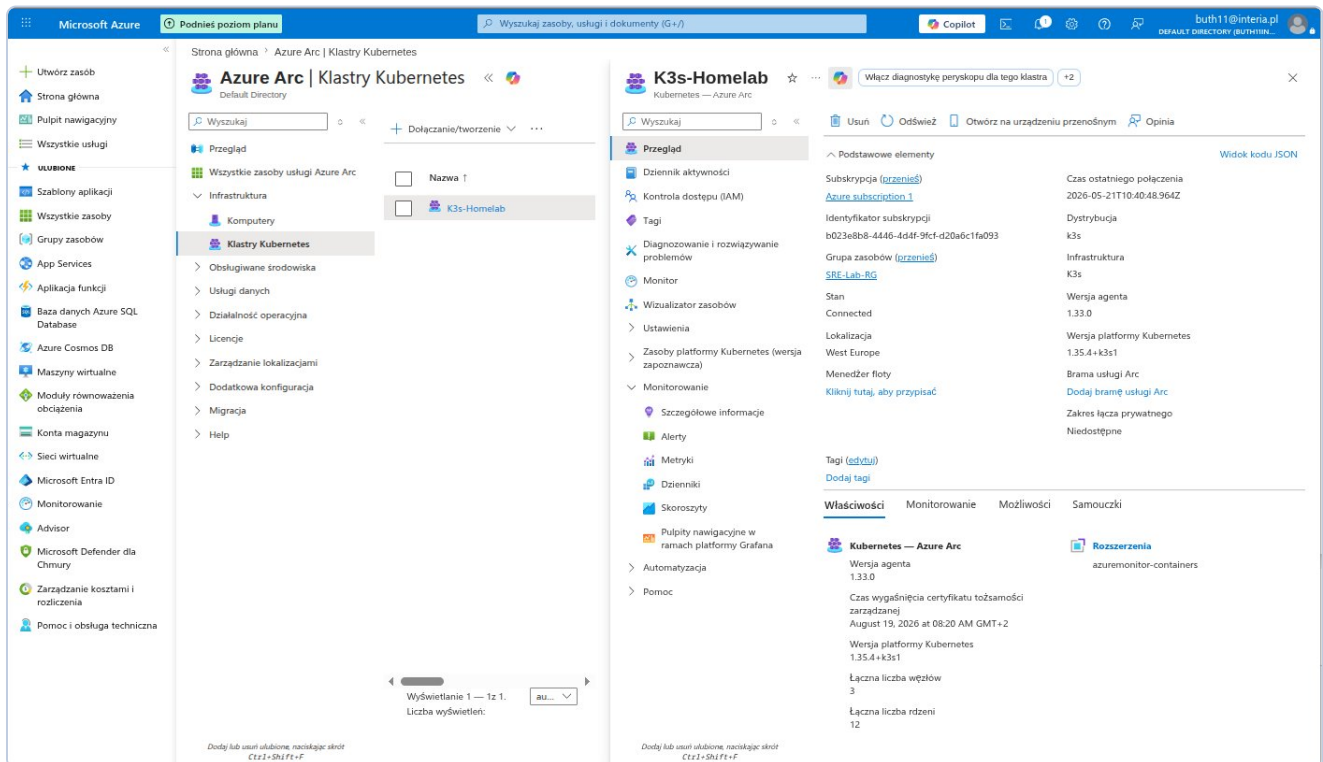
Rozdział 3: FinOps

The screenshot displays the Microsoft Azure Cost Management dashboard for user Bartosz Suszko. The interface is in Polish. On the left, there is a navigation pane with various Azure services and cost management tools. The main area shows the 'Budżety' (Budgets) section. A table lists the budget details for 'SRE-Lab-Budget'.

Nazwa	Zakres	Resetuj okres	Data utworzenia	Data wygaśnięcia	Budżet	Prognostyczne	Oszacowane wy...	Postęp
SRE-Lab-Budget	55df28b8-39ed-58b...	Monthly	1.05.2026	30.04.2028	5,00 USD		\$0.00	0.00%

Zrzut 1: Azure Cost Management — budżet 5 USD/mies., koszt: 0.00 USD.

Rozdział 4: Azure Arc



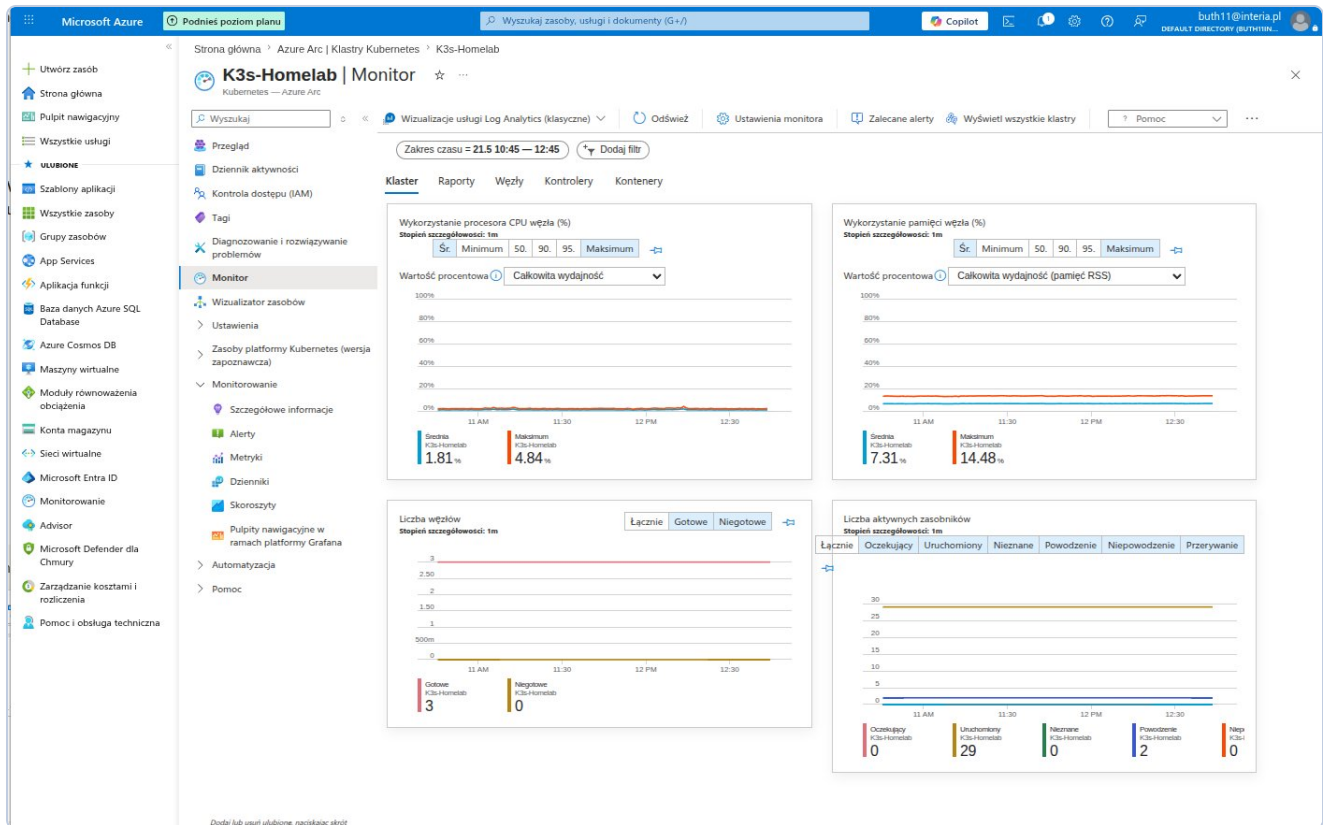
The screenshot displays the Microsoft Azure portal interface. The top navigation bar shows the user's profile (buth11@interia.pl) and the 'Podnieś poziom planu' button. The left sidebar contains the navigation menu with categories like 'Utwórz zasób', 'Strona główna', 'Pulpit nawigacyjny', and 'Wszystkie usługi'. The main content area is titled 'Azure Arc | Klastry Kubernetes'. It shows a list of clusters, with 'K3s-Homelab' selected. The cluster details page for 'K3s-Homelab' is displayed, showing its status as 'Connected' and its version as '1.35.4+k3s1'. The page also shows the number of nodes (3) and the total number of CPUs (12). The 'Właściwości' (Properties) section lists various attributes of the cluster, including its name, location, and status. The 'Monitorowanie' (Monitoring) section shows the cluster's health and performance metrics.

Zrzut 2: Azure Arc — K3s-Homelab Connected, v1.35.4+k3s1, 3 węzły, 12 rdzeni.

Rozdział 5: Observability

```
Desktop: bash × (buth11) 192.168.122.10 × (buth11) 192.168.122.10 × (buth11) 192.168.122.10 × Desktop: bash × (buth11) 192.168.122.10 ×
{
  "principalId": "a3fe4a8c-7a2e-4b99-8de9-2150d617cb76",
  "tenantId": null,
  "type": "SystemAssigned"
},
"isSystemExtension": false,
"name": "azuremonitor-containers",
"packageUri": null,
"plan": null,
"provisioningState": "Succeeded",
"releaseTrain": "Stable",
"resourceGroup": "SRE-Lab-RG",
"scope": {
  "cluster": {
    "releaseNamespace": "azuremonitor-containers"
  },
  "namespace": null
},
"statuses": [],
"systemData": {
  "createdAt": "2026-05-22T07:44:41.936626+00:00",
  "createdBy": null,
  "createdByType": null,
  "lastModifiedAt": "2026-05-22T07:44:41.936626+00:00",
  "lastModifiedBy": null,
  "lastModifiedByType": null
},
"type": "Microsoft.KubernetesConfiguration/extensions",
"version": null
}
buth11@k3s-master:~$ sudo k3s kubectl get pods -n kube-system | grep ama
[sudo] password for buth11:
ama-logs-9m9x9          3/3      Running    0          37s
ama-logs-9scpr          3/3      Running    0          37s
ama-logs-rs-65b58c7f96-f8lds 2/2      Running    0          37s
ama-logs-spcdp          3/3      Running    0          37s
buth11@k3s-master:~$
```

Zrzut 3: Pody AMA Running — 4 agenty, 0 restartow.



Zrzut 4: Azure Monitor — CPU avg 1.58%, RAM avg 6.27%, 3 wezly Ready, 29 podow.

Rozdział 6: KQL

The screenshot displays the Microsoft Azure portal interface. On the left is the navigation pane with 'Log Analytics' selected. The main area shows the 'Logs' section for a specific workspace. A KQL query is entered and executed, showing a table of results.

Query:

```
1 KubeNodeInventory
2 where TimeGenerated > ago(1h)
3 summarize arg_max(TimeGenerated, *) by Computer
4 project
5   Computer,
6   Status,
7   KubeletVersion
8   order by Computer asc
```

Results Table:

Computer	Status	KubeletVersion
k3s-master	Ready	v1.35.4+k3s1
k3s-worker1	Ready	v1.35.4+k3s1
k3s-worker2	Ready	v1.35.4+k3s1

KQL-1: 3 nodes Ready, v1.35.4+k3s1.

DefaultWorkspace-b023e8b8-4446-4d4f-9fcf-d20a6c1fa093-WEU | Logs

Log Analytics workspace

Search

New Query 1* ... +

Time range: Set in query Show: 1000 results KQL mode

```

1 KubePodInventory
2 where TimeGenerated > ago(1h)
3 summarize arg_max(TimeGenerated, *) by Name, Namespace
4 summarize PodCount = count() by Namespace, PodStatus
5 order by Namespace asc, PodCount desc

```

Namespace	PodStatus	PodCount
azure-arc	Running	12
default	Running	3
kube-system	Running	14
kube-system	Succeeded	2

0s 533ms Display time (UTC+00:00) Query details 1 - 4 of 4

KQL-2: azure-arc 12 Running, default 3 Running, kube-system 14+2.

DefaultWorkspace-b023e8b8-4446-4d4f-9fcf-d20a6c1fa093-WEU | Logs

Log Analytics workspace

Search

New Query 1* ... +

Time range: Set in query Show: 1000 results KQL mode

```

1 KubeEvents
2 where TimeGenerated > ago(24h)
3 project TimeGenerated, Namespace, Name, Reason, Message
4 order by TimeGenerated desc
5 take 20

```

TimeGenerated [UTC]	Namespace	Name	Reason
5/22/2026, 7:44:56.000 AM	kube-system	ama-logs-9cpr	FailedMount
5/22/2026, 7:44:56.000 AM	kube-system	ama-logs-rs-65b58c7f96-f8ld	FailedMount
5/22/2026, 7:35:32.000 AM	azure-arc	kube-aad-proxy-5f59465457-12...	Unhealthy
5/22/2026, 7:35:25.000 AM	azure-arc	config-agent-7548f59c5-k4zrv	Unhealthy
5/22/2026, 7:35:13.000 AM	k3s-worker2		Rebooted
5/22/2026, 7:35:11.000 AM	k3s-worker1		Rebooted
5/22/2026, 7:35:09.000 AM	kube-system	metrics-server-786d997795-zz...	Unhealthy
5/22/2026, 7:35:07.000 AM	k3s-master		Rebooted
5/21/2026, 10:53:39.000 AM	kube-system	ama-logs-89tpj	Unhealthy
5/21/2026, 10:52:38.000 AM	kube-system	ama-logs-rs-65b58c7f96-nkshw	Unhealthy
5/21/2026, 8:37:39.000 AM	kube-system	svclb-aras-demo-app-6f6f7352...	FailedScheduling
5/21/2026, 8:37:39.000 AM	kube-system	svclb-aras-demo-app-6f6f7352...	FailedScheduling
5/21/2026, 8:37:39.000 AM	kube-system	svclb-aras-demo-app-6f6f7352...	FailedScheduling

0s 287ms Display time (UTC+00:00) Query details 1 - 13 of 13

KQL-3: 13 eventow 24h — Rebooted, Unhealthy (chwilowe), FailedScheduling.

The screenshot shows the Microsoft Azure Log Analytics interface. The left sidebar contains navigation options like Home, Dashboard, All services, and Favorites. The main pane displays a KQL query for pod restarts in the last 24 hours. The query filters for pods with a restart count greater than 0 and orders them by restart count in descending order. The results table shows various pods, including kube-system pods like kube-arc, kube-system, and kube-arc, all with a restart count of 2 and a status of Running.

Namespace	Name	PodRestartCount	PodStatus
kube-arc	extension-manager-85899588c...	3	Running
kube-arc	clusterconnect-agent-5bc68597...	3	Running
kube-system	svclb-traefik-4b75cb6-8b8x...	2	Running
kube-arc	extension-events-collector-6c9...	2	Running
kube-arc	clusteridentityoperator-767768...	2	Running
kube-arc	kube-aad-proxy-5f5465457-42...	2	Running
kube-system	svclb-traefik-4b75cb6-c2gzs...	2	Running
kube-arc	config-agent-7554859c5-k4zrv...	2	Running
kube-arc	cluster-metadata-operator-6db...	2	Running
kube-system	svclb-traefik-4b75cb6-9587...	2	Running
kube-arc	metrics-agent-6d994764-rc44...	2	Running
kube-arc	controller-manager-7cb948ddff...	2	Running
kube-arc	resource-sync-agent-5f4688cc5...	2	Running
kube-arc	flux-logs-agent-76f8c454c9-9g...	1	Running
kube-arc	logcollector-8bd4468b4-4s2zw...	1	Running
kube-system	ama-logs-nlg5p	1	Running
kube-system	helm-install-traefik-ggts...	1	Succeeded
kube-system	svclb-aras-demo-app-8581d79...	1	Running
default	aras-demo-app-f78659546-vb...	1	Running
kube-system	coredns-c4d8fbb5f-llkx...	1	Running
kube-system	local-path-provisioner-5c4dc5d...	1	Running
kube-system	metrics-server-786d997795-zz...	1	Running
kube-system	svclb-aras-demo-app-8581d79...	1	Running

KQL-4: 27 podow z restartami po restarcie VM — wszystkie Running.

The screenshot shows the Microsoft Azure Log Analytics interface. The left sidebar contains navigation options like Home, Dashboard, All services, and Favorites. The main pane displays a KQL query for pod availability in the last 24 hours. The query filters for pods with a status of 'Ready' and calculates the availability percentage. The results table shows three computers: k3s-master, k3s-worker1, and k3s-worker2, all with an availability percentage of 100 and 184 ready samples out of 184 total samples.

Computer	AvailabilityPct	ReadySamples	TotalSamples
k3s-master	100	184	184
k3s-worker1	100	184	184
k3s-worker2	100	184	184

KQL-5: SLO 100% — 184/184 probek Ready.

SLO 100% w oknie 24h. Error budget nienaruszony nawet po pelnym restarcie VM.

Rozdział 7: Load Balancing

```
Desktop: bash × (buth11) 192.168.122.10 × (buth11) 192.168.122.10 × (buth11) 192.168.122.10 × Desktop: bash × (buth11) 192.168.122.10 ×
https://ubuntu.com/engage/secure-kubernetes-at-the-edge

Expanded Security Maintenance for Applications is not enabled.

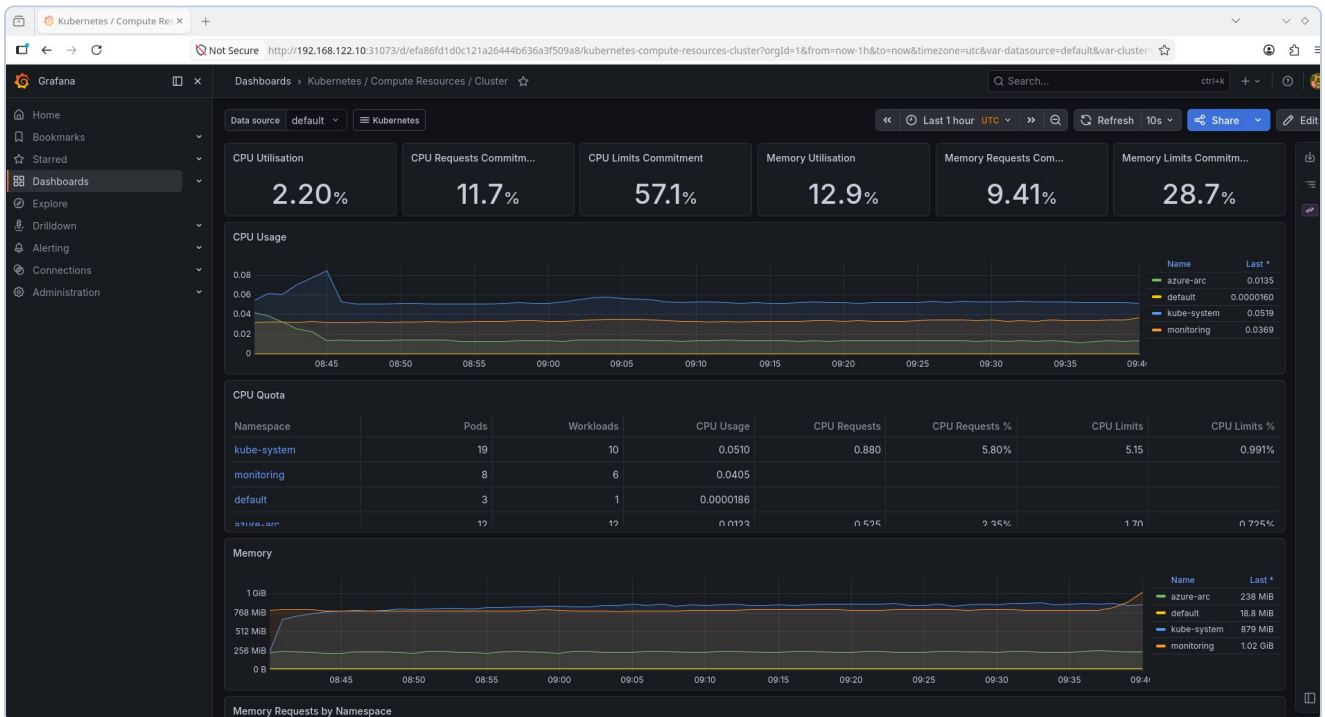
0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Fri May 22 07:42:36 2026 from 192.168.122.1
buth11@k3s-master:~$ for i in {1..20}; do
  curl -s http://192.168.122.10:8080 | grep "Hostname"
  sleep 0.5
done
Hostname: aras-demo-app-f78659546-d7h9c
Hostname: aras-demo-app-f78659546-d7h9c
Hostname: aras-demo-app-f78659546-4lprm
Hostname: aras-demo-app-f78659546-4lprm
Hostname: aras-demo-app-f78659546-d7h9c
Hostname: aras-demo-app-f78659546-4lprm
Hostname: aras-demo-app-f78659546-d7h9c
Hostname: aras-demo-app-f78659546-d7h9c
Hostname: aras-demo-app-f78659546-4lprm
Hostname: aras-demo-app-f78659546-d7h9c
Hostname: aras-demo-app-f78659546-4lprm
Hostname: aras-demo-app-f78659546-d7h9c
Hostname: aras-demo-app-f78659546-vbwkm
Hostname: aras-demo-app-f78659546-vbwkm
Hostname: aras-demo-app-f78659546-vbwkm
Hostname: aras-demo-app-f78659546-4lprm
Hostname: aras-demo-app-f78659546-vbwkm
Hostname: aras-demo-app-f78659546-4lprm
Hostname: aras-demo-app-f78659546-4lprm
Hostname: aras-demo-app-f78659546-4lprm
buth11@k3s-master:~$
```

Zrzut 5: Round-robin — 20 requestow na 3 pody, 900 rownolegla bez bledow.

Rozdział 8: Prometheus & Grafana



Rzrzut 6: Grafana — CPU 2.20%, RAM 12.9%, CPU Quota per namespace.

Rozdział 9: Terraform IaC

```
Desktop: bash × (buth11) 192.168.122.10 × (buth11) 192.168.122.10 × (buth11) 192.168.122.10 × terraform: bash × (buth11) 192.168.122.10 ×
Plan: 6 to add, 0 to change, 0 to destroy.

Changes to Outputs:
+ action_group_id           = (known after apply)
+ budget_name               = "SRE-Lab-Budget"
+ log_analytics_workspace_id = (known after apply)
+ log_analytics_workspace_key = (sensitive value)
+ resource_group_id         = (known after apply)
+ resource_group_name        = "SRE-Lab-RG"
+ subscription_id           = "b023e8b8-4446-4d4f-9fcf-d20a6c1fa093"

Saved the plan to: tfplan

To perform exactly these actions, run the following command to apply:
  terraform apply "tfplan"
buth11@fedora:~/VMs/Azure/Github_repo/sre_repo/terraform$ terraform apply tfplan
random_string.suffix: Creating...
random_string.suffix: Creation complete after 0s [id=5plwbb]
azurerm_resource_group.sre_lab: Creating...
azurerm_consumption_budget_subscription.sre_lab: Creating...
azurerm_consumption_budget_subscription.sre_lab: Still creating... [00m10s elapsed]
azurerm_consumption_budget_subscription.sre_lab: Creation complete after 19s [id=/subscriptions/b023e8b8-4446-4d4f-9fcf-d20a6c1fa093/providers/Microsoft.Consumption/budgets/SRE-Lab-Budget]

Error: A resource with the ID "/subscriptions/b023e8b8-4446-4d4f-9fcf-d20a6c1fa093/resourceGroups/SRE-Lab-RG" already exists -
to be managed via Terraform this resource needs to be imported into the State. Please see the resource documentation for "azure
rm_resource_group" for more information.

with azurerm_resource_group.sre_lab,
on main.tf line 26, in resource "azurerm_resource_group" "sre_lab":
26: resource "azurerm_resource_group" "sre_lab" {

buth11@fedora:~/VMs/Azure/Github_repo/sre_repo/terraform$
```

Terraform Plan: 6 zasobow, 0 zmian.

```
Desktop: bash × (buth11) 192.168.122.10 × (buth11) 192.168.122.10 × (buth11) 192.168.122.10 × terraform : bash × (buth11) 192.168.122.10 ×

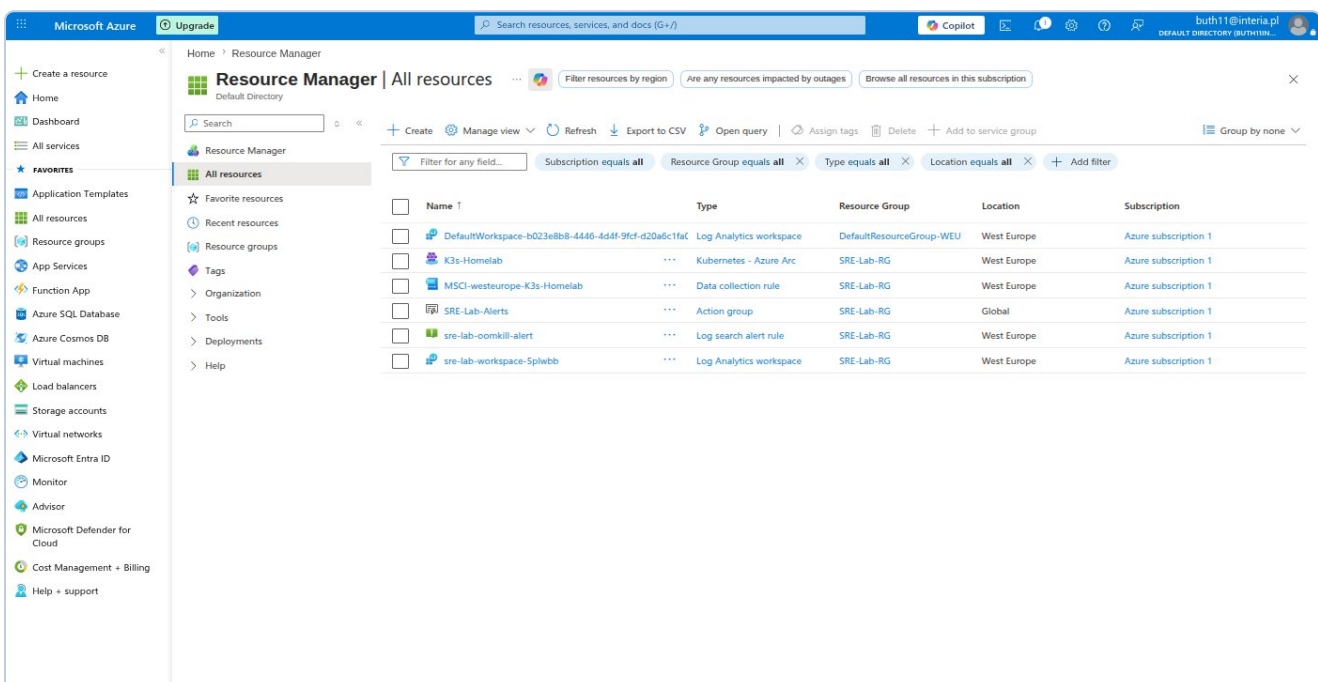
Saved the plan to: tfplan

To perform exactly these actions, run the following command to apply:
terraform apply "tfplan"
azurerm_resource_group.sre_lab: Modifying... [id=/subscriptions/b023e8b8-4446-4d4f-9fcf-d20a6c1fa093/resourceGroups/SRE-Lab-RG]
azurerm_resource_group.sre_lab: Modifications complete after 3s [id=/subscriptions/b023e8b8-4446-4d4f-9fcf-d20a6c1fa093/resourceGroups/SRE-Lab-RG]
azurerm_monitor_action_group.sre_alerts: Creating...
azurerm_log_analytics_workspace.sre_lab: Creating...
azurerm_monitor_action_group.sre_alerts: Creation complete after 4s [id=/subscriptions/b023e8b8-4446-4d4f-9fcf-d20a6c1fa093/resourceGroups/SRE-Lab-RG/providers/Microsoft.Insights/actionGroups/SRE-Lab-Alerts]
azurerm_log_analytics_workspace.sre_lab: Still creating... [00m10s elapsed]
azurerm_log_analytics_workspace.sre_lab: Still creating... [00m20s elapsed]
azurerm_log_analytics_workspace.sre_lab: Still creating... [00m30s elapsed]
azurerm_log_analytics_workspace.sre_lab: Still creating... [00m40s elapsed]
azurerm_log_analytics_workspace.sre_lab: Creation complete after 47s [id=/subscriptions/b023e8b8-4446-4d4f-9fcf-d20a6c1fa093/resourceGroups/SRE-Lab-RG/providers/Microsoft.OperationalInsights/workspaces/sre-lab-workspace-5plwbb]
azurerm_monitor_scheduled_query_rules_alert_v2.oom_kill: Creating...
azurerm_monitor_scheduled_query_rules_alert_v2.oom_kill: Creation complete after 5s [id=/subscriptions/b023e8b8-4446-4d4f-9fcf-d20a6c1fa093/resourceGroups/SRE-Lab-RG/providers/Microsoft.Insights/scheduledQueryRules/sre-lab-oomkill-alert]

Apply complete! Resources: 3 added, 1 changed, 0 destroyed.

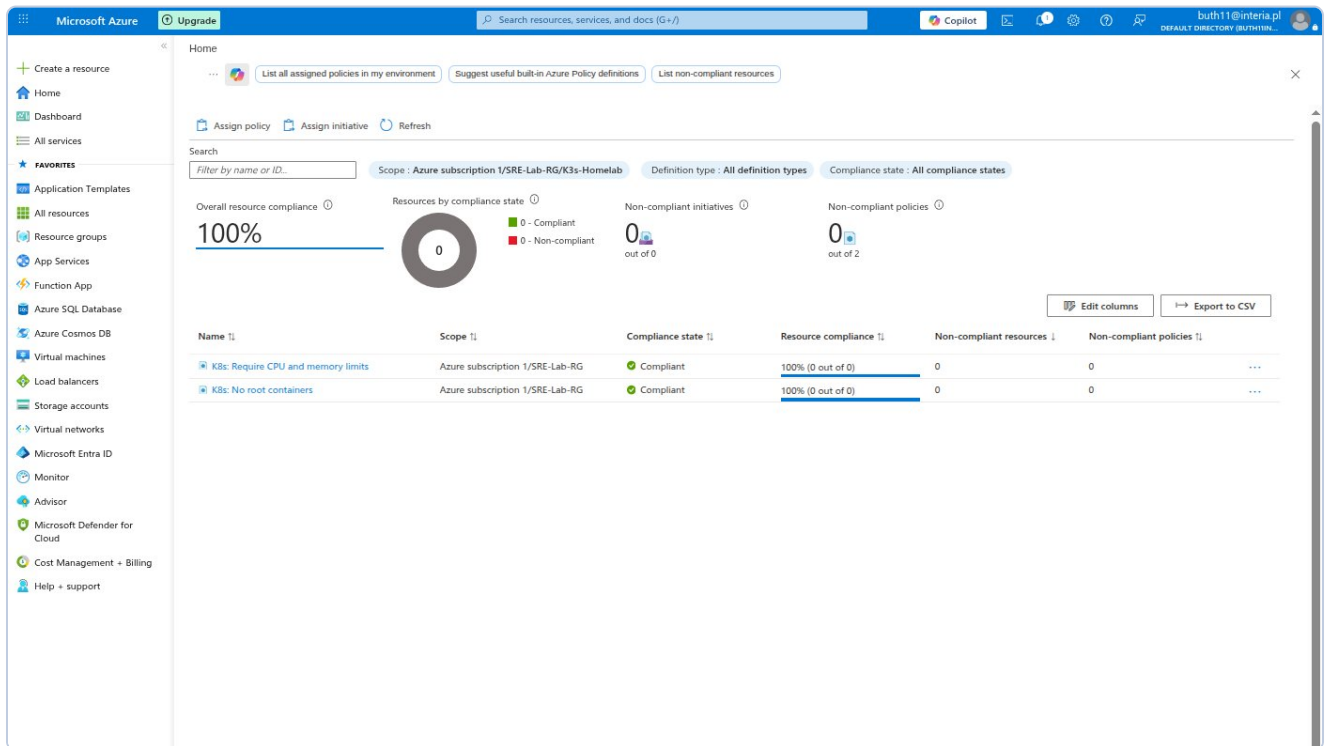
Outputs:
action_group_id = "/subscriptions/b023e8b8-4446-4d4f-9fcf-d20a6c1fa093/resourceGroups/SRE-Lab-RG/providers/Microsoft.Insights/actionGroups/SRE-Lab-Alerts"
budget_name = "SRE-Lab-Budget"
log_analytics_workspace_id = "/subscriptions/b023e8b8-4446-4d4f-9fcf-d20a6c1fa093/resourceGroups/SRE-Lab-RG/providers/Microsoft.OperationalInsights/workspaces/sre-lab-workspace-5plwbb"
log_analytics_workspace_key = <sensitive>
resource_id = "/subscriptions/b023e8b8-4446-4d4f-9fcf-d20a6c1fa093/resourceGroups/SRE-Lab-RG"
resource_group_name = "SRE-Lab-RG"
subscription_id = "b023e8b8-4446-4d4f-9fcf-d20a6c1fa093"
buth11@fedora:~/VMs/Azure/Github_repo/sre_repo/terraform$
```

Terraform Apply: 3 dodane, 1 zmieniony.



Portal Azure: zasoby Terraform w SRE-Lab-RG.

Rozdział 10: Azure Policy



Azure Policy: No root containers + CPU/memory limits — oba Compliant.

Rozdział 11: GitHub

The screenshot shows the GitHub repository page for 'buth11 / sre-homelab-azure-arc'. The repository is public and has 1 branch (main) and 0 tags. The file list shows the initial commit: 'K3s + Azure Arc SRE Homelab' 8 minutes ago. The files listed are: docs, k8s, monitoring, scripts, terraform, .gitignore, and README.md. The README file is selected, showing the title 'SRE Homelab: Azure Arc Hybrid Cloud Lab' and a description: 'A Proof-of-Value project demonstrating hybrid cloud orchestration, infrastructure-as-code, observability, and FinOps practices using K3s Kubernetes and Microsoft Azure Arc.' The right sidebar shows the repository's statistics: 0 stars, 0 watching, and 0 forks. The 'About' section is empty. The 'Releases' section shows no releases published. The 'Packages' section shows no packages published. The 'Contributors' section shows 1 contributor: buth11. The 'Languages' section shows a bar chart with Shell at 58.4% and HCL at 41.6%.

GitHub repository page for **buth11 / sre-homelab-azure-arc** (Public).

Repository statistics: 1 Branch, 0 Tags, 0 Stars, 0 Watching, 0 Forks.

Initial commit: K3s + Azure Arc SRE Homelab (3d8283c · 8 minutes ago) · 1 Commit

Files in repository:

- docs
- k8s
- monitoring
- scripts
- terraform
- .gitignore
- README.md

README content:

SRE Homelab: Azure Arc Hybrid Cloud Lab

A Proof-of-Value project demonstrating hybrid cloud orchestration, infrastructure-as-code, observability, and FinOps practices using K3s Kubernetes and Microsoft Azure Arc.

Overview

Contributors: 1 (buth11)

Languages: Shell 58.4%, HCL 41.6%

GitHub repo sre-homelab-azure-arc — publiczne, Shell 58.4%, HCL 41.6%.

Rozdział 12: Czesc 2 — Service Principal, Remote State, CI/CD

Remote State — 14.53 KiB, szyfrowanie, wersjonowanie. State lock przy kazdym plan.

Pipeline DevOps: validate + plan (37s) → Approval Gate → apply (49s) = 4m 24s total.

Microsoft Azure Upgrade Search resources, services, and docs (G+/I) buth11@interia.pl DEFAULT DIRECTORY (BUTH11IN...

Create a resource

Home Dashboard All services FAVORITES Application Templates All resources Resource groups App Services Function App Azure SQL Database Azure Cosmos DB Virtual machines Load balancers Storage accounts Virtual networks Microsoft Entra ID Monitor Advisor Microsoft Defender for Cloud Cost Management + Billing Help + support

sre-homelab.tfstate

Blob

Save Discard Download Refresh Delete Change tier Acquire lease Break lease Give feedback

Overview Versions Snapshots Edit Generate SAS

Properties

URL	https://sreftfstate5496.blob...
LAST MODIFIED	5/23/2026, 8:23:07 PM
CREATION TIME	5/23/2026, 8:23:07 PM
VERSION ID	2026-05-23T18:23:07.9861361Z
TYPE	Block blob
SIZE	14.53 KiB
ACCESS TIER	Hot (Inferred)
ACCESS TIER LAST MODIFIED	N/A
ARCHIVE STATUS	-
REHYDRATE PRIORITY	-
SERVER ENCRYPTED	true
ETAG	0x8DEB8F856BD8961
VERSION-LEVEL IMMUTABILITY POLICY	Disabled
CACHE-CONTROL	
CONTENT-TYPE	application/json
CONTENT-MD5	2mHfbTY53lmCzqo/RW5+MQ...
CONTENT-ENCODING	
CONTENT-LANGUAGE	
CONTENT-DISPOSITION	
LEASE STATUS	Unlocked
LEASE STATE	Available
LEASE DURATION	-
COPY STATUS	-
COPY COMPLETION TIME	-

Undelete

Metadata

Key	Value

Blob index tags

Key	Value

Zrzut W2-2: Portal Azure — blob properties: 14.53 KiB, encrypted, versioning active.

```
buth11@fedora:~/VMs/Azure/Github_repo/sre_repo/terraform$ # terraform plan pobierze state z Azure Blob zamiast lokalnego pliku
# Powinno pokazać "No changes" bo infrastruktura jest aktualna
terraform plan
Acquiring state lock. This may take a few moments...
random_string.suffix: Refreshing state... [id=5plwbb]
data.azurearm_subscription.current: Reading...
azurearm_resource_group.sre_lab: Refreshing state... [id=/subscriptions/b023e8b8-4446-4d4f-9fcf-d20a6c1fa093/resourceGroups/SRE-Lab-RG]
azurearm_monitor_action_group.sre_alerts: Refreshing state... [id=/subscriptions/b023e8b8-4446-4d4f-9fcf-d20a6c1fa093/resourceGroups/SRE-Lab-RG/providers/Microsoft.Insights/actionGroups/SRE-Lab-Alerts]
azurearm_log_analytics_workspace.sre_lab: Refreshing state... [id=/subscriptions/b023e8b8-4446-4d4f-9fcf-d20a6c1fa093/resourceGroups/SRE-Lab-RG/providers/Microsoft.OperationalInsights/workspaces/sre-lab-workspace-5plwbb]
data.azurearm_subscription.current: Read complete after 1s [id=/subscriptions/b023e8b8-4446-4d4f-9fcf-d20a6c1fa093]
azurearm_consumption_budget_subscription.sre_lab: Refreshing state... [id=/subscriptions/b023e8b8-4446-4d4f-9fcf-d20a6c1fa093/providers/Microsoft.Consumption/budgets/SRE-Lab-Budget]
azurearm_monitor_scheduled_query_rules_alert_v2.oom_kill: Refreshing state... [id=/subscriptions/b023e8b8-4446-4d4f-9fcf-d20a6c1fa093/resourceGroups/SRE-Lab-RG/providers/Microsoft.Insights/scheduledQueryRules/sre-lab-oomkill-alert]

No changes. Your infrastructure matches the configuration.

Terraform has compared your real infrastructure against your configuration and found no differences, so no changes are needed.
Releasing state lock. This may take a few moments...
buth11@fedora:~/VMs/Azure/Github_repo/sre_repo/terraform$
```

Zrzut W2-3: terraform plan — Acquiring/Releasing state lock. No changes.

The screenshot shows the Azure DevOps web interface. On the left is a sidebar with navigation options: Overview, Boards, Repos, Pipelines, Environments, Releases, Library (selected), Task groups, Deployment groups, Test Plans, and Artifacts. The main area is titled 'Library > terraform-credentials'. Below this, there are tabs for 'Variable group', 'Save', 'Clone', 'Security', 'Pipeline permissions', 'Approvals and checks', and 'Help'. The 'Variable group' tab is active, showing the 'Properties' section with 'Variable group name' set to 'terraform-credentials' and an empty 'Description' field. There is a toggle switch for 'Link secrets from an Azure key vault as variables' which is currently turned off. Below the properties is the 'Variables' section, which contains a table with the following data:

Name ↑	Value	
ARM_CLIENT_ID	54c7e017-a745-4e1e-b9da-d74f4901a3de	
ARM_CLIENT_SECRET	*****	
ARM_SUBSCRIPTION_ID	b023e8b8-4446-4d4f-9fcf-d20a6c1fa093	
ARM_TENANT_ID	fd7288c4-e003-40fd-b3cf-9c0d1821cf5d	

At the bottom of the variables section, there is a '+ Add' button. The bottom of the interface shows 'Project settings' and a back arrow.

Zrzut W2-4: Azure DevOps Variable Group — ARM_* z ukrytym CLIENT_SECRET.

Azure DevOps

buth11

/ sre-homelab-arc

/ Pipelines

/ Environments

/ production

Search

BS

sre-homelab-arc

+

Overview

Boards

Repos

Pipelines

Pipelines

Environments

Releases

Library

Task groups

Deployment groups

Test Plans

Artifacts

Project settings

<<

production

Automatically created environment

Add resource

Deployments

Approvals and checks

Check execution order

Checks will run by order, which is pre-determined by check types. Checks within the same order will run in parallel.
[Learn more](#)

+ Add new

Order	Display name	Type	Timeout	Details
1	All approvers must approve	Approvals	30d	8%

Zrzut W2-5: Environment production — Approval Gate aktywna.

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Azure DevOps Interface showing a pipeline run for **sre-homelab-arc** with ID **#20260523.1**.

Jobs in run

Job Name	Status	Duration
terraform plan	Success	34s
Initialize job	Success	<1s
Checkout buth1...	Success	2s
Install Terraform...	Success	2s
terraform init	Success	7s
terraform validate	Success	1s
terraform plan	Success	15s
Publish terrafor...	Success	3s
Post-job: Chec...	Success	<1s
Finalize Job	Success	<1s

Terraform Validate & Plan

Terraform Apply

- terraform apply

terraform plan Log:

```
1 Agent: Azure Pipelines 1
2 Started: Just now
3 Duration: 34s
4
5 Job preparation parameters
44 1 artifact produced
45 Job live console data:
46 Starting: terraform plan
47 Async Command Start: DetectDockerContainer
48 Async Command End: DetectDockerContainer
49 Async Command Start: DetectDockerContainer
50 Async Command End: DetectDockerContainer
51 Finishing: terraform plan
```

Zrzut W2-6: Stage 1 — wszystkie kroki zielone, 1 artifact.

Azure DevOps | sre-homelab-arc | Pipelines | #20260523.1 • feat: add Azure DevOps CI/CD pipeline for Terraform

Run new

This run is being retained as one of 3 recent runs by pipeline. View retention leases

Summary | Environments | Associated pipelines | Code Coverage

Manually run by Bartosz Suszko
Today at 9:13 PM 4m 24s

Repository and version: sre-homelab-arc / main e9da4bfb

Related: 0 work items, 1 published

Tests and coverage: Get started

View change | Parameters

Stages | Jobs

Terraform Validate ...
1 job completed 37s
1 artifact

Terraform Apply
1 job completed 49s
1 checks passed

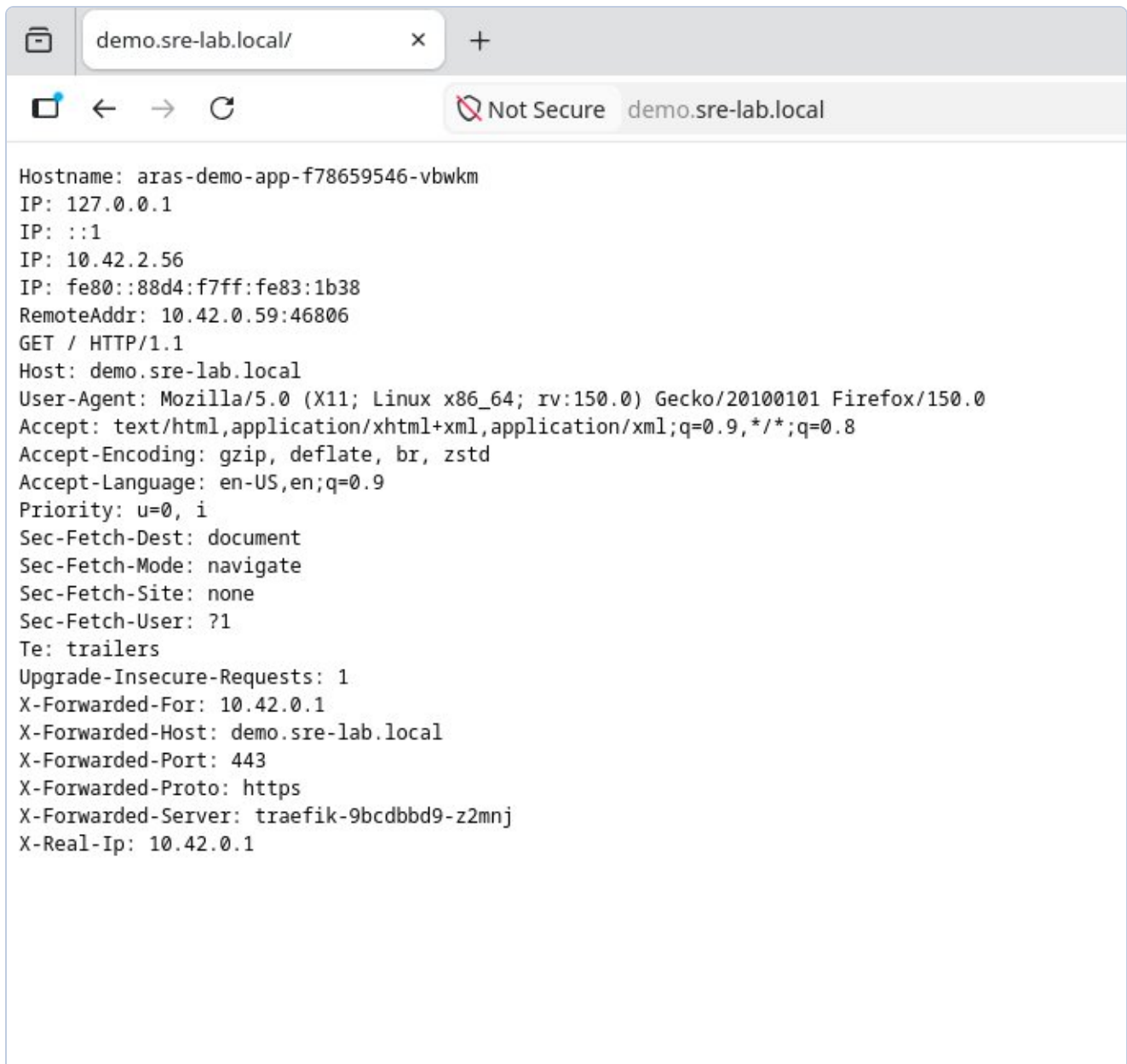
Project settings

Zrzut W2-7: Pipeline kompletny — oba stage zielone, 4m 24s.

Rozdział 13: Część 3 — Zaawansowane operacje Kubernetes

CZĘŚĆ 3: Ingress TLS, HPA, NetworkPolicy, RBAC

Traefik HTTPS, self-signed TLS, HPA 2-8 replik (860% CPU), NetworkPolicy deny-all, RBAC least privilege.



Zrzut W3-1: Przeglądarka Firefox — <https://demo.sre-lab.local> działa. Widoczne: Hostname poda, X-Forwarded-Proto: https, X-Forwarded-Server: traefik-9bcd9bd9-z2mnj, X-Forwarded-Port: 443.

HPA — automatyczne skalowanie na podstawie CPU

Horizontal Pod Autoscaler skonfigurowany dla Deployment aras-demo-app. HPA co 15 sekund odpytuje metrics-server o zużycie CPU. Gdy średnie CPU Podów przekroczy 50% — dodaje repliki. Gdy spada poniżej 50% przez 5 minut — usuwa.

- Zakres: minReplicas: 2, maxReplicas: 8, targetCPU: 50%
- Test obciążeniowy: `ab -n 50000 -c 50` — 1621 req/s przez 30 sekund
- Scale-up: CPU 860% → automatyczne dodanie replik do 8
- Scale-down: CPU 6% po 5 minutach → powrót do 2 replik (minimum)

Every 5.0s: sudo k3s kubectl get hpa &... k3s-master: Sun May 24 09:32:49 2026

NAME	REFERENCE	TARGETS	MINPODS	MAXPODS
aras-demo-hpa	Deployment/aras-demo-app	cpu: 860%/50%	2	8
REPLICAS	AGE			
4	9m55s			

aras-demo-app-5b58b48d6d-5vwl4	1/1	Running 0	24s	
aras-demo-app-5b58b48d6d-7mbln	1/1	Running 0	10s	
aras-demo-app-5b58b48d6d-859c7	1/1	Running 0	6m32s	
aras-demo-app-5b58b48d6d-96qs4	1/1	Running 0	10s	
aras-demo-app-5b58b48d6d-phz5d	1/1	Running 0	6m27s	
aras-demo-app-5b58b48d6d-sbv4q	1/1	Running 0	10s	
aras-demo-app-5b58b48d6d-xx94p	1/1	Running 0	24s	
aras-demo-app-5b58b48d6d-zrmgs	1/1	Running 0	10s	

Zrzut W3-2: HPA scale-up — cpu: 860%/50%, REPLICAS: 8 (max), 8 podow Running. Automatyczne skalowanie w akcji podczas testu obciążeniowego ab.

```
Every 5.0s: sudo k3s kubectl get hpa &... k3s-master: Sun May 24 09:38:17 2026

NAME          REFERENCE          TARGETS          MINPODS  MAXPODS  RE
PLICAS    AGE
aras-demo-hpa  Deployment/aras-demo-app  cpu: 6%/50%    2         8         8
15m
---
aras-demo-app-5b58b48d6d-phz5d  1/1    Running    0         11m
aras-demo-app-5b58b48d6d-zrmgs  1/1    Running    0         5m38s
```

Zrzut W3-3: HPA scale-down — cpu: 6%/50%, REPLICAS: 2 (min). Po 5 minutach bez ruchu HPA automatycznie usunął 6 zbędnych Podów.

NetworkPolicy — firewall wewnątrz klastra

Domyslnie wszystkie Pody w Kubernetes mogą się komunikować ze wszystkimi. NetworkPolicy definiuje reguły jak firewall — który namespace i które Pody mogą łączyć się z aplikacją demo. Wdrożono politykę blokującą cały ruch przychodzący z wyjątkiem dozwolonych źródeł.

- Zezwolone: kube-system (Traefik), monitoring (Prometheus), repliki aras-demo-app
- Zablokowane: wszystko inne
- Weryfikacja: nieautoryzowany Pod — timeout/blokada. Traefik — odpowiedź aplikacji

RBAC — least privilege dla aplikacji

Role-Based Access Control — każda aplikacja powinna mieć własne konto (ServiceAccount) z dokładnie tymi uprawnieniami których potrzebuje. Zasada least privilege: jeśli aplikacja nie potrzebuje kasować Podów, nie powinna mieć tego robić nawet jeśli zostanie przejęta przez atakującego.

- ServiceAccount: aras-demo-sa (zamiast wspólnego default)
- Role: get/list/watch na Podach i ConfigMapach — nic więcej
- Weryfikacja: can-i list pods = yes, can-i delete pods = no

```

buth11@k3s-master:~$ # Test 1: nieautoryzowany Pod probuje polaczyc sie z aplik
acja
# timeout 5 = czekamy max 5 sekund na odpowiedz
# Oczekiwany wynik: brak odpowiedzi / timeout - polityka blokuje
echo "=== Test: nieautoryzowany dostep (powinien byc zablokowany) ==="
sudo k3s kubectl run test-blocked \
  --image=curlimages/curl \
  --rm -it \
  --restart=Never \
  -- curl -s --max-time 5 http://aras-demo-app:8080 || echo "ZABLOKOWANY - Netw
orkPolicy dziala"

# Test 2: sprawdzamy ze Ingress (Traefik) nadal dziala
# Traefik jest w kube-system wiec ma dostep
echo "=== Test: dostep przez Ingress (powinien dzialac) ==="
curl -sk https://demo.sre-lab.local | head -2
=== Test: nieautoryzowany dostep (powinien byc zablokowany) ===
pod "test-blocked" deleted from default namespace
pod default/test-blocked terminated (Error)
ZABLOKOWANY - NetworkPolicy dziala
=== Test: dostep przez Ingress (powinien dzialac) ===
buth11@k3s-master:~$ # Test dostępu przez Traefik - uzywamy IP zamiast domeny
# Traefik nasluchuje na porcie 80 i 443 na wszystkich wezlach
# -H podajemy naglowek Host zeby Traefik wiedzial do ktorego serwisu kierowac
echo "=== Test dostep przez Traefik (kube-system) ==="
curl -s -H "Host: demo.sre-lab.local" http://192.168.122.10 | head -3

# Dodatkowo sprawdzamy ze bezposredni dostep do portu 8080
# przez serwis nadal dziala (LoadBalancer Service jest poza NetworkPolicy)
echo "=== Test dostep przez LoadBalancer Service ==="
curl -s http://192.168.122.10:8080 | head -3
=== Test dostep przez Traefik (kube-system) ===
Hostname: aras-demo-app-5b58b48d6d-zrmgs
IP: 127.0.0.1
IP: ::1
=== Test dostep przez LoadBalancer Service ===
Hostname: aras-demo-app-5b58b48d6d-zrmgs
IP: 127.0.0.1
IP: ::1
buth11@k3s-master:~$

```

Zrzut W3-4: RBAC weryfikacja — ServiceAccount aras-demo-sa: list pods=yes (dozwolone), delete pods=no (zablokowane). Zasada least privilege potwierdzona.

Rozdział 14: Postmortem POST-001

Pole	Wartosc
Severity	P2
Czas	Ponizej 7 minut
Root Cause	Literowka: ks3-worker1 zamiast k3s-worker1
Status	Rozwiazany — regex validation w skryptach

Rozdział 15: Wyniki i wnioski (Czesci 1-3)

Metryka	Czesc 1	Czesc 2	Czesc 3
Wezly	3/3 Ready	—	—
SLO	100%	—	—
Koszty	\$0.00	\$0.00	\$0.00
Terraform state	Local	Remote (Blob)	—
CI/CD	—	DevOps 4m24s	—
Ingress TLS	—	—	HTTPS verified
HPA	—	—	3 na 8 na 2
NetworkPolicy	—	—	Zablokowane
RBAC	—	—	Least privilege
Incydenty	6	8	10

Rozdział 16: Troubleshooting Log

ISSUE-001 KQL: CPUCapacityNanoCores column missing

project operator: Failed to resolve scalar expression named CPUCapacityNanoCores

Rozwiązanie: KubeNodeInventory | getschema — uproszczono zapytanie do dostępnych kolumn.

ISSUE-002 Azure Policy: MissingPolicyParameter cpuLimit/memoryLimit

(MissingPolicyParameter) policy assignment k8s-resource-limits is missing cpuLimit, memoryLimit

Rozwiązanie: Dodano --params z cpuLimit: 500m i memoryLimit: 512Mi.

ISSUE-003 Terraform: Resource Group already exists (brownfield)

Error: A resource with the ID .../SRE-Lab-RG already exists

Rozwiązanie: terraform import azurerm_resource_group.sre_lab /subscriptions/.../SRE-Lab-RG

ISSUE-004 Terraform: Saved plan is stale after import

Error: Saved plan is stale – state was changed after plan was created

Rozwiązanie: Nowy terraform plan -out=tfplan po imporcie.

ISSUE-005 Container Insights missing after VM restart

Rozwiązanie: az k8s-extension create --name azuremonitor-containers — agenty wrocily w 37s.

ISSUE-006 POST-001: etcd Split-Brain (ks3 vs k3s)

FATA[0034] Node password rejected, duplicate hostname

Rozwiązanie: kubectl delete node + rm /etc/rancher/node/ + restart k3s-agent. Skrypty zaktualizowane o regex validation.

ISSUE-007 Azure DevOps: TerraformInstaller task missing

A task is missing: TerraformInstaller

Rozwiązanie: Instalacja z Marketplace: ms-devlabs.custom-terraform-tasks.

ISSUE-008 Pipeline: permission for Variable Group

This pipeline needs permission to access a resource

Rozwiązanie: View → Permit — jednorazowe zatwierdzenie.

ISSUE-009 sudo does not expand ~ in paths

error: Cannot read file ~/ingress/server.crt, open ~/ingress/server.crt: no such file or directory

Przyczyna: sudo uruchamia komendę jako root który ma inny katalog domowy. Tylda ~ rozwija się do /root/ zamiast /home/buth11/.

Rozwiązanie: Użyć pełnej ścieżki: /home/buth11/ingress/server.crt zamiast ~/ingress/server.crt

ISSUE-010 HPA shows cpu: unknown — missing resource requests

```
NAME TARGETS MINPODS MAXPODS REPLICAS — cpu: <unknown>/50%
```

Przyczyna: HPA oblicza procent CPU jako $\text{aktualne_zuzycie}/\text{requests} \times 100$. Bez zdefiniowanych requests w Deployment, HPA nie ma od czego liczyć procentów.

Rozwiązanie: Dodano resources.requests do Deployment: cpu: 10m, memory: 32Mi. Po rolling update HPA zaczął pokazywać poprawne wartości: cpu: 0%/50%.

Rozdział 17: Część 4 — GitOps z FluxCD 2.8.8

FluxCD bootstrapowany z repozytorium GitHub przez klucz SSH. Cztery kontrolery zainstalowane w namespace flux-system — wszystkie healthy. Kustomization gitops-demo synchronizuje namespace gitops-demo.

```

buth11@fedora: ~/VMs/Azure/Github_repo/sre_repo$ # Part 4
flux get all && kubectl get pods -n flux-system && kubectl get pods -n mdc

# Part 5
kubectl get clusterissuer && kubectl get certificate -n default && curl -sk https://demo.sre-lab.local | grep -E "Hostname|X-Forwarded"
NAME                                REVISION          SUSPENDED    READY    MESSAGE
gitrepository/flux-system           main@sha1:2658d5e5 False        True     stored artifact for revision 'main@sha1:2658d5e5'

NAME                                REVISION          SUSPENDED    READY    MESSAGE
kustomization/flux-system           main@sha1:2658d5e5 False        True     Applied revision: main@sha1:2658d5e5
kustomization/gitops-demo           main@sha1:2658d5e5 False        True     Applied revision: main@sha1:2658d5e5

NAME                                READY    STATUS    RESTARTS    AGE
helm-controller-5984cc88cf-94jkw    1/1      Running   1 (121m ago) 2d3h
kustomize-controller-d6bb746df-9khwf 1/1      Running   1 (121m ago) 2d3h
notification-controller-7bc5c97f8d-jdflb 1/1      Running   1 (121m ago) 2d3h
source-controller-745c67ff9-2g8qd    1/1      Running   1 (121m ago) 2d3h
NAME                                READY    STATUS    RESTARTS    AGE
microsoft-defender-collectors-ds-cn56h 2/2      Running   2 (121m ago) 24h
microsoft-defender-collectors-ds-pz7cf 2/2      Running   2 (121m ago) 24h
microsoft-defender-collectors-ds-tfq8b 2/2      Running   2 (121m ago) 24h
microsoft-defender-pod-collector-misc-5d785599b4-zr2xw 1/1      Running   1 (121m ago) 24h
microsoft-defender-publisher-ds-5d6cs 1/1      Running   1 (121m ago) 24h
microsoft-defender-publisher-ds-7qgjm 1/1      Running   1 (121m ago) 24h
microsoft-defender-publisher-ds-hkq5m 1/1      Running   1 (121m ago) 24h
NAME                                READY    AGE
selfsigned-issuer                    True     119m
sre-lab-ca-issuer                     True     119m
NAME                                READY    SECRET    AGE
demo-sre-lab-tls                     True     demo-sre-lab-tls 118m
Hostname: aras-demo-app-5b58b48d6d-spwkm
X-Forwarded-For: 10.42.0.1
X-Forwarded-Host: demo.sre-lab.local
X-Forwarded-Port: 443
X-Forwarded-Proto: https
X-Forwarded-Server: traefik-9bcd9bd9-z2mnj
buth11@fedora: ~/VMs/Azure/Github_repo/sre_repo$

```

Zrzut W4-1: flux get all — gitrepository/flux-system i kustomization/flux-system + kustomization/gitops-demo, wszystkie Ready: True. kubectl get pods -n flux-system — 4 kontrolery Running. GitOps test: zmiana replicas 2→3 przez git push, trzecia replika pojawiła się automatycznie.

Rozdział 18: Czesc 4 — Trivy — skanowanie CVE

Trivy 0.70.0 przeskanował obraz traefik/whoami:latest. Wszystkie podatnosci dotycza stdlib Go v1.24.1.

```
buth11@fedora:~/VMs/Azure/Github_repo/sre_repo$ trivy image traefik/whoami:latest --severity CRITICAL,HIGH --format table 2>/dev/null | grep -A3 "CRITICAL\|Report Summary"
Report Summary

+-----+-----+-----+-----+-----+-----+-----+
| Target | Type | Vulnerabilities | Secrets |
+-----+-----+-----+-----+-----+-----+-----+
--
Total: 30 (HIGH: 29, CRITICAL: 1)

+-----+-----+-----+-----+-----+-----+-----+
| Library | Vulnerability | Severity | Status | Installed Version | Fixed Version | Title |
+-----+-----+-----+-----+-----+-----+-----+
| stdlib | CVE-2025-68121 | CRITICAL | fixed | v1.24.1 | 1.24.13, 1.25.7, 1.26.0-rc.3 | crypto/tls: crypto/tls: Incorrect certificate validation during TLS session resumption https://avd.aquasec.com/nvd/cve-2025-68121 |
+-----+-----+-----+-----+-----+-----+-----+
```

Zrzut W4-2: Trivy Report Summary — Total: 30 (HIGH: 29, CRITICAL: 1). CVE-2025-68121 CRITICAL — crypto/tls: nieprawidlowa walidacja certyfikatu. Fixed: Go 1.24.13, 1.25.7.

```
buth11@fedora:~/VMs/Azure/Github_repo/sre_repo$ trivy image traefik/whoami:latest --severity CRITICAL,HIGH --format table 2>/dev/null | tail -20
+-----+-----+-----+-----+-----+-----+-----+
| CVE-2026-39823 | | | | | CVE-2026-27142 fixed a vulnerability in which URLs were not correctly ..... https://avd.aquasec.com/nvd/cve-2026-39823 |
+-----+-----+-----+-----+-----+-----+-----+
| CVE-2026-39825 | | | | | ReverseProxy can forward queries containing parameters not visible to ... https://avd.aquasec.com/nvd/cve-2026-39825 |
+-----+-----+-----+-----+-----+-----+-----+
| CVE-2026-39826 | | | | | If a trusted template author were to write a <script> tag containing... https://avd.aquasec.com/nvd/cve-2026-39826 |
+-----+-----+-----+-----+-----+-----+-----+
| CVE-2026-39836 | | | | | Panic in Dial and LookupPort when handling NUL byte on Windows in... https://avd.aquasec.com/nvd/cve-2026-39836 |
+-----+-----+-----+-----+-----+-----+-----+
| CVE-2026-42499 | | | | | Pathological inputs could cause DoS through consumePhrase when parsing ... https://avd.aquasec.com/nvd/cve-2026-42499 |
+-----+-----+-----+-----+-----+-----+-----+
```

Zrzut W4-3: Trivy — lista CVE HIGH (fragment). CVE-2026-39823, CVE-2026-39825 (ReverseProxy), CVE-2026-39826 (XSS), CVE-2026-39836 (NUL panic), CVE-2026-42499 (DoS).

Rozdział 19: Część 4 — Defender for Cloud

Defender for Containers włączony na poziomie Standard (30-dniowy trial). 7 podów DaemonSet w namespace mdc — po jednym na każdym węzle.

Zrzut W4-1 (powyżej, sekcja `kubectl get pods -n mdc`) pokazuje wyniki: `microsoft-defender-collectors-ds-*` (3x, 2/2 Running), `microsoft-defender-publisher-ds-*` (3x, 1/1 Running), `microsoft-defender-pod-collector-misc` (1/1 Running).

Rozdział 20: Czesć 5 — cert-manager v1.20.2

cert-manager zainstalowany przez Helm. Automatyczne wystawianie i odnawianie certyfikatów TLS. Certyfikat demo-sre-lab-tls wystawiony w 22 sekundy przez adnotację Ingress.

```
buth11@fedora:~/VMs/Azure/Github_repo/sre_repo$ # Part 4
flux get all && kubectl get pods -n flux-system && kubectl get pods -n mdc

# Part 5
kubectl get clusterissuer && kubectl get certificate -n default && curl -sk https://demo.sre-lab.local | grep -E "Hostname|X-Forwarded"
NAME                                REVISION    SUSPENDED    READY    MESSAGE
gitrepository/flux-system           main@sha1:2658d5e5    False       True     stored artifact for revision 'main@sha1:2658d5e5'

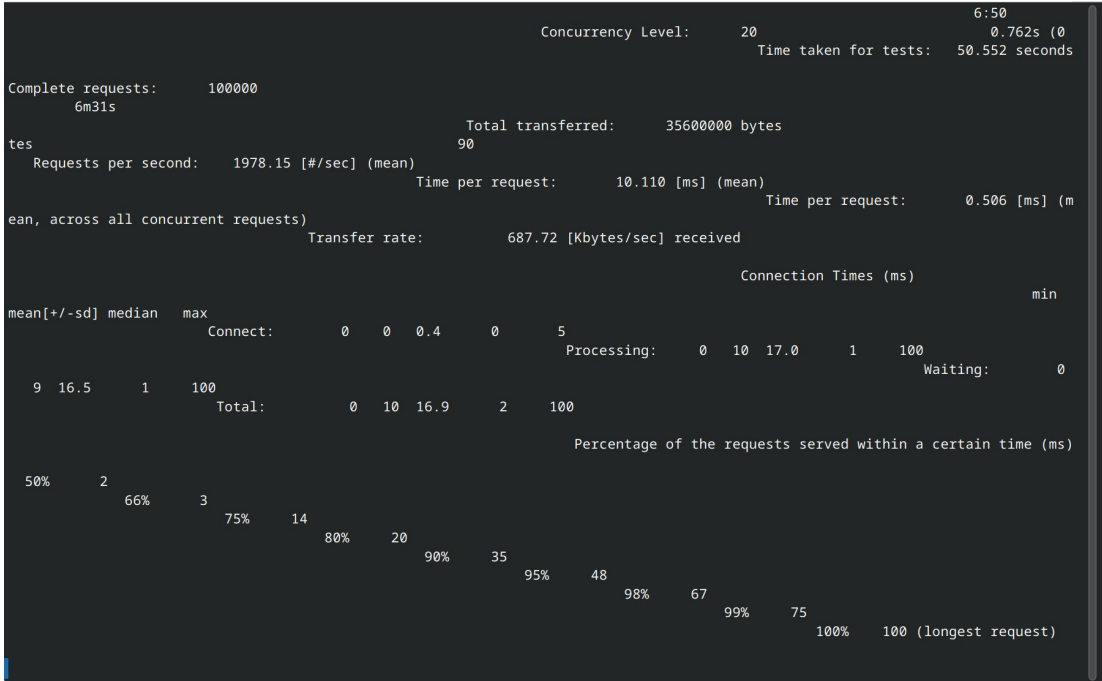
NAME                                REVISION    SUSPENDED    READY    MESSAGE
kustomization/flux-system           main@sha1:2658d5e5    False       True     Applied revision: main@sha1:2658d5e5
kustomization/gitops-demo           main@sha1:2658d5e5    False       True     Applied revision: main@sha1:2658d5e5

NAME                                READY    STATUS    RESTARTS    AGE
helm-controller-5984cc88cf-94jkw    1/1      Running   1 (121m ago)  2d3h
kustomize-controller-d6bb746df-9khwf 1/1      Running   1 (121m ago)  2d3h
notification-controller-7bc5c97f8d-jdf1b 1/1      Running   1 (121m ago)  2d3h
source-controller-745c67ff9-2g8qd    1/1      Running   1 (121m ago)  2d3h
NAME                                READY    STATUS    RESTARTS    AGE
microsoft-defender-collectors-ds-cn56h 2/2      Running   2 (121m ago)  24h
microsoft-defender-collectors-ds-pz7cf 2/2      Running   2 (121m ago)  24h
microsoft-defender-collectors-ds-tfq8b 2/2      Running   2 (121m ago)  24h
microsoft-defender-pod-collector-misc-5d785599b4-zr2xw 1/1      Running   1 (121m ago)  24h
microsoft-defender-publisher-ds-5d6cs 1/1      Running   1 (121m ago)  24h
microsoft-defender-publisher-ds-7qgjm 1/1      Running   1 (121m ago)  24h
microsoft-defender-publisher-ds-hkq5m 1/1      Running   1 (121m ago)  24h
NAME                                READY    AGE
selfsigned-issuer                   True     119m
sre-lab-ca-issuer                   True     119m
NAME                                READY    SECRET    AGE
demo-sre-lab-tls                   True     demo-sre-lab-tls 118m
Hostname: aras-demo-app-5b58b48d6d-spwkm
X-Forwarded-For: 10.42.0.1
X-Forwarded-Host: demo.sre-lab.local
X-Forwarded-Port: 443
X-Forwarded-Proto: https
X-Forwarded-Server: traefik-9bcd9bd9-z2mnj
buth11@fedora:~/VMs/Azure/Github_repo/sre_repo$
```

Zrzut W5-1: kubectl get clusterissuer — selfsigned-issuer True, sre-lab-ca-issuer True. kubectl get certificate -n default — demo-sre-lab-tls Ready: True, AGE: 118m. curl weryfikacja: X-Forwarded-Proto: https, X-Forwarded-Port: 443, X-Forwarded-Server: traefik.

Rozdział 21: Czesc 6 — KEDA Event-Driven Autoscaling

KEDA skaluje aplikacje na podstawie metryk Prometheusa (container_cpu_usage_seconds_total). ScaledObject: minReplicaCount=1, maxReplicaCount=8, cooldownPeriod=30s.



Zrzut W6-1: ab baseline — 100000 requestow, 1978 req/s, 50.552s.

```
Every 2.0s: kubectl get scaledobject -n default && echo '---' && kubectl get pods -n default | ... fedora: Wed 27 May 2026 12:52:21 PM CEST
in 0.575s (0)
NAME          SCALETARGETKIND  SCALETARGETNAME  MIN  MAX  READY  ACTIVE  FALLBACK  PAUSED  TRIGGERS  AUTHENTICATIONS  A
GE
aras-demo-keda  apps/v1.Deployment  aras-demo-app    1    8    True   True    False    False    prometheus                                     2
2m
---
aras-demo-app-5b58b48d6d-spwkm  1/1    Running  0      7m46s
```

Zrzut W6-2: KEDA ACTIVE: True — scale-up uruchomiony. 2 repliki Running.

```
Every 2.0s: kubectl get scaledobject -n default && echo '---' && kubectl get pods -n default | ... fedora: Wed 27 May 2026 12:53:04 PM CEST
in 0.998s (0)
NAME          SCALETARGETKIND  SCALETARGETNAME  MIN  MAX  READY  ACTIVE  FALLBACK  PAUSED  TRIGGERS  AUTHENTICATIONS  A
GE
aras-demo-keda  apps/v1.Deployment  aras-demo-app    1    8    True   False   False    False   prometheus                2
2m
---
aras-demo-app-5b58b48d6d-dhfhb  1/1    Running  0      13s
aras-demo-app-5b58b48d6d-q9qtz  1/1    Running  0      13s
aras-demo-app-5b58b48d6d-spwkm  1/1    Running  0      8m29s
aras-demo-app-5b58b48d6d-zrtlx  1/1    Running  0      13s
```

Zrzut W6-3: KEDA scale-up — 4 repliki Running.

```
Every 2.0s: kubectl get scaledobject -n default && echo '---' && kubectl get pods -n default | ... fedora: Wed 27 May 2026 12:53:17 PM CEST
in 0.210s (0)
NAME          SCALETARGETKIND  SCALETARGETNAME  MIN  MAX  READY  ACTIVE  FALLBACK  PAUSED  TRIGGERS  AUTHENTICATIONS  A
GE
aras-demo-keda  apps/v1.Deployment  aras-demo-app    1    8    True   False   False    False   prometheus                2
2m
---
aras-demo-app-5b58b48d6d-2q7pp  1/1    Running  0      11s
aras-demo-app-5b58b48d6d-82m24  1/1    Running  0      11s
aras-demo-app-5b58b48d6d-99s8c  1/1    Running  0      11s
aras-demo-app-5b58b48d6d-dhfhb  1/1    Running  0      26s
aras-demo-app-5b58b48d6d-ftqmw  1/1    Running  0      11s
aras-demo-app-5b58b48d6d-q9qtz  1/1    Running  0      26s
aras-demo-app-5b58b48d6d-spwkm  1/1    Running  0      8m42s
aras-demo-app-5b58b48d6d-zrtlx  1/1    Running  0      26s
```

Zrzut W6-4: KEDA scale-up szczyt — 8 replik Running (maxReplicaCount osiagniety).

```

80% 19
90% 34
95% 43
98% 53
99% 61
100% 147 (longest request)
buth11@fedora:~/VMs/Azure/Github_repo/sre_repo$ ssh buth11@192.168.122.10 "ab -n 200000 -c 50 http://localhost:8080/"
This is ApacheBench, Version 2.3 <$Revision: 1903618 $>
Copyright 1996 Adam Twiss, Zeus Technology Ltd, http://www.zeustech.net/
Licensed to The Apache Software Foundation, http://www.apache.org/

Benchmarking localhost (be patient)
Completed 20000 requests
Completed 40000 requests
Completed 60000 requests
Completed 80000 requests
Completed 100000 requests
Completed 120000 requests
Completed 140000 requests
Completed 160000 requests
Completed 180000 requests
Completed 200000 requests
Finished 200000 requests

Server Software:
Server Hostname: localhost
Server Port: 8080

Document Path: /
Document Length: 238 bytes

Concurrency Level: 50
Time taken for tests: 72.781 seconds
Complete requests: 200000
Failed requests: 121082
  (Connect: 0, Receive: 0, Length: 121082, Exceptions: 0)
Total transferred: 71226416 bytes
HTML transferred: 47626416 bytes
Requests per second: 2747.95 [#/sec] (mean)
Time per request: 18.195 [ms] (mean)
Time per request: 0.364 [ms] (mean, across all concurrent requests)
Transfer rate: 955.70 [Kbytes/sec] received

Connection Times (ms)
      min      mean[+/-sd] median   max
Connect:    0     2   1.4      1     9
Processing:  0    16  29.0      4   495
Waiting:    0    15  28.4      3   494
Total:       0    18  29.0      7   496

Percentage of the requests served within a certain time (ms)
 50%    7
 66%   11
 75%   20
 80%   26
 90%   48
 95%   87
 98%  100
 99%  105
100%  496 (longest request)
buth11@fedora:~/VMs/Azure/Github_repo/sre_repo$

```

Zrzut W6-5: ab 200000 requestow — 2747 req/s, 72.781s.

```

Every 2.0s: kubectl get scaledobject -n default && echo '---' && kubectl get pods -n default | ... fedora: Wed 27 May 2026 12:59:49 PM CEST
in 0.163s (0)
NAME          SCALETARGETKIND  SCALETARGETNAME  MIN  MAX  READY  ACTIVE  FALLBACK  PAUSED  TRIGGERS  AUTHENTICATIONS  A
GE
aras-demo-keda  apps/v1.Deployment  aras-demo-app    1    8    True   False   False     False   prometheus                2
---
aras-demo-app-5b58b48d6d-spwkm  1/1    Running    0    15m

```

Zrzut W6-6: KEDA scale-down — ACTIVE: False, 1 replika. Cooldown 30s minal.

Rozdział 22: Wyniki i wnioski (Czesci 1-6)

Metryka	Cz.1	Cz.2	Cz.3	Cz.4	Cz.5	Cz.6
Wezly	3/3 Ready	—	—	—	—	—
SLO	100%	—	—	—	—	—
Koszty	\$0.00	\$0.00	\$0.00	\$0.00	\$0.00	\$0.00
CI/CD	—	DevOps 4m24s	—	—	—	—
Ingress TLS	—	—	Reczny	—	cert-manager	—
Autoscaling	—	—	HPA 2-8	—	—	KEDA 1-8
Scale to zero	—	—	Nie	—	—	Tak
Cooldown	—	—	5 min	—	—	30 sek
GitOps	—	—	—	FluxCD	—	—
CVE scan	—	—	—	1 CRIT+12 HIGH	—	—
Runtime sec.	—	—	—	Defender	—	—
Auto TLS	—	—	—	—	cert-manager	—
Incydenty	6	8	10	11	12	13

Rozdział 23: Troubleshooting Log (13 incydentów)

ISSUE-001 — KQL: CPUCapacityNanoCores column missing

Rozwiązanie: Użyj getschema do weryfikacji dostępnych kolumn

ISSUE-002 — Azure Policy: brak parametrow cpuLimit/memoryLimit

Rozwiązanie: Dodaj --params do az policy assignment create

ISSUE-003 — Terraform: Resource Group już istnieje

Rozwiązanie: terraform import azurerm_resource_group.sre_lab

ISSUE-004 — Terraform: stary plan po imporcie

Rozwiązanie: Wygeneruj nowy terraform plan -out=tfplan

ISSUE-005 — Container Insights brak po restarcie VM

Rozwiązanie: az k8s-extension create --name azuremonitor-containers

ISSUE-006 — POST-001: etcd Split-Brain (ks3 vs k3s)

Rozwiązanie: kubectl delete node + rm /etc/rancher/node/ + restart k3s-agent

ISSUE-007 — Azure DevOps: brak TerraformInstaller

Rozwiązanie: Instalacja z Marketplace: ms-devlabs.custom-terraform-tasks

ISSUE-008 — Pipeline: brak uprawnień do Variable Group

Rozwiązanie: View → Permit — jednorazowe zatwierdzenie

ISSUE-009 — sudo nie rozwija ~ w ścieżkach

Rozwiązanie: Użyj pełnej ścieżki /home/buth11/ zamiast ~/

ISSUE-010 — HPA pokazuje cpu: unknown

Rozwiązanie: Dodaj resources.requests do Deployment: cpu: 10m

ISSUE-011 — RBAC RoleBinding: apiRef zamiast apiGroup

Rozwiązanie: Popraw pole na apiGroup: rbac.authorization.k8s.io

ISSUE-012 — cert-manager webhook: duration < 1h odrzucone

Rozwiązanie: Minimum duration: 1h5m, renewBefore: 6m

ISSUE-013 — KEDA: konflikt z istniejącym HPA

Rozwiązanie: Usun HPA przed wdrożeniem ScaledObject

CZESC 7

Helm — własny chart aras-demo-app

Własny chart Helm v0.1.0 dla aplikacji aras-demo-app. Templates: Deployment, Service, Ingress (cert-manager), RBAC, NetworkPolicy, NOTES.txt.

Zademonstrowali: helm install, helm upgrade --set, helm rollback, helm history.

=== HELM ===

NAME	NAMESPACE	REVISION	UPDATED	STATUS	CHAR	
T	APP VERSION					
aras-demo	default	3	2026-06-06 17:02:26.737603427 +0200 CEST	deployed	aras	
-demo-app-0.1.0	1.0.0					
REVISION	UPDATED		STATUS	CHART	APP VERSION	DESCRIPTION
1	Sat Jun 6 17:00:57 2026		superseded	aras-demo-app-0.1.0	1.0.0	Install comp
lete						
2	Sat Jun 6 17:01:56 2026		superseded	aras-demo-app-0.1.0	1.0.0	Upgrade comp
lete						
3	Sat Jun 6 17:02:26 2026		deployed	aras-demo-app-0.1.0	1.0.0	Rollback to
1						

Zrzut W7-1: helm list — aras-demo deployed REVISION 3 (po rollback). helm history — REVISION 1 Install, REVISION 2 Upgrade (3 repliki), REVISION 3 Rollback to 1.

```
=== PODS ===
```

aras-demo-6ff6ffd5b-dqigz	2/2	Running	0	27m
aras-demo-6ff6ffd5b-fq6fg	2/2	Running	0	27m
aras-demo-app-5b58b48d6d-spwkm	1/1	Running	1 (90m ago)	10d

Zrzut W7-2: kubectl get pods — 2x aras-demo 2/2 Running (Linkerd sidecar), 1x stary pod 1/1.

CZESC 8

OPA/Gatekeeper — Policy as Code Audit

Gatekeeper zarządzany przez Azure Arc — 16 ConstraintTemplates od Azure Policy. Tryb dryrun — audyt naruszeń bez blokowania. Własne ConstraintTemplates są blokowane przez webhook Azure.

Polityka	Naruszenia	Tryb
k8sazurev3containerlimits	34	dryrun
k8sazurev2containerallowedimages	30	dryrun
k8sazurev2blockautomounttoken	21	dryrun
k8sazurev3allowedusersgroups	22	dryrun
k8sazurev2noprivilege	0	dryrun
k8sazurev3disallowedcapabilities	0	dryrun

NAME	ENFORCEMENT
NT-ACTION TOTAL-VIOLATIONS	
NAME	E
NFORCEMENT-ACTION TOTAL-VIOLATIONS	
NAME	
ENFORCEMENT-ACTION TOTAL-VIOLATIONS	
NAME	
ENFORCEMENT-ACTION TOTAL-VIOLATIONS	
NAME	
ENFORCEMENT-ACTION TOTAL-VIOLATIONS	
NAME	
ENFORCEMENT-ACTION TOTAL-VIOLATIONS	
NAME	ENFORCEMENT
-ACTION TOTAL-VIOLATIONS	
k8sazurev2noprivilege.constraints.gatekeeper.sh/azurepolicy-k8sazurev2noprivilege-37aeb6fdd17c74c0451c	dryrun
3	
k8sazurev2noprivilege.constraints.gatekeeper.sh/azurepolicy-k8sazurev2noprivilege-ccc4cfa94c3dfff6cbc1	dryrun
3	
NAME	
ENFORCEMENT-ACTION TOTAL-VIOLATIONS	

Zrzut W8-1: `kubectl get constraints` — `k8sazurev2noprivilege`: `dryrun`, 0 naruszeń (aplikacja zgodna z polityką `no-privilege`). Pełna lista 16 `ConstraintTemplates` z Azure Policy.

CZESC 9

Chaos Engineering — Chaos Mesh

Chaos Mesh zainstalowany z konfiguracją dla K3s containerd. Eksperyment PodChaos pod-kill co 30 sekund. Self-healing zweryfikowany — nowe Pody pojawiają się w ciągu 3-5 sekund.

```
=== CHAOS MESH ===
NAME                                READY   STATUS    RESTARTS   AGE
chaos-controller-manager-85cbb588b9-25hcn 1/1     Running   0           67m
chaos-controller-manager-85cbb588b9-7wr8b 1/1     Running   0           67m
chaos-controller-manager-85cbb588b9-1mj4z 1/1     Running   0           67m
chaos-daemon-btxgv                     1/1     Running   0           67m
chaos-daemon-rprt6                     1/1     Running   0           67m
chaos-daemon-wd9cx                     1/1     Running   0           67m
chaos-dashboard-5b77585c6c-2ngtg        1/1     Running   0           67m
chaos-dns-server-6bccb5bb76-g7xtw       1/1     Running   0           67m
=== LINKERD ===
NAME      MESHED   SUCCESS   RPS   LATENCY P50   LATENCY P95   LATENCY P99   TCP_CONN
```

Zrzut W9-1: `kubectl get pods -n chaos-mesh` — 8 podów Running: 3x chaos-controller-manager, 3x chaos-daemon (DaemonSet — jeden na każdy węzeł), chaos-dashboard, chaos-dns-server.

```
Every 2.0s: kubectl get scaledobject -n default && echo '---' && kubectl get pods -n default | ... fedora: Wed 27 May 2026 12:53:17 PM CEST
in 0.210s (0)
NAME          SCALETARGETKIND  SCALETARGETNAME  MIN  MAX  READY  ACTIVE  FALLBACK  PAUSED  TRIGGERS  AUTHENTICATIONS  A
GE
aras-demo-keda apps/v1.Deployment aras-demo-app    1    8    True   False   False     False   prometheus  2
2m
---
aras-demo-app-5b58b48d6d-2q7pp 1/1    Running   0      11s
aras-demo-app-5b58b48d6d-82m24 1/1    Running   0      11s
aras-demo-app-5b58b48d6d-99s8c 1/1    Running   0      11s
aras-demo-app-5b58b48d6d-dhfhb 1/1    Running   0      26s
aras-demo-app-5b58b48d6d-ftqmw 1/1    Running   0      11s
aras-demo-app-5b58b48d6d-q9qtz 1/1    Running   0      26s
aras-demo-app-5b58b48d6d-spwkm 1/1    Running   0      8m42s
aras-demo-app-5b58b48d6d-zrtlx 1/1    Running   0      26s
```

Zrzut W9-2: Podczas eksperymentu PodChaos — 8 podów aras-demo Running jednocześnie. Kubernetes natychmiast zastąpił zabite pody nowymi.

CZESC 10

Linkerd Service Mesh — mTLS

Linkerd edge-26.6.1 zainstalowany z viz extension. Namespace default annotowany — sidecar proxy wstrzykniety automatycznie do wszystkich nowych Podow. MESHED: 2/2.

```
aras-demo      2/2      -      -      -      -      -      -  
=== OPA ===  
NAME                                                    ENFORCEMENT  
NT-ACTION      TOTAL-VIOLATIONS
```

Zrzut W10-1: linkerd viz stat deployment/aras-demo — MESHED: 2/2. Oba pody w service mesh z aktywnym mTLS.

```
=== PODS ===  
aras-demo-6ff6ffd5b-dqrgz      2/2      Running      0          27m  
aras-demo-6ff6ffd5b-fq6fg      2/2      Running      0          27m  
aras-demo-app-5b58b48d6d-spwkm 1/1      Running      1 (90m ago) 10d
```

Zrzut W10-2: kubectl get pods -n default — aras-demo 2/2 Running (2 kontenery: aplikacja whoami + linkerd-proxy sidecar). Stary pod aras-demo-app 1/1 bez sidecar.

Podsumowanie wyników — Części 1-10

Metryka	P1	P2	P3	P4	P5	P6	P7	P8	P9	P10
Koszty	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0
SLO	100%	—	—	—	—	—	—	—	—	—
CI/CD	—	4m24s	—	—	—	—	—	—	—	—
TLS	—	—	Reczny	—	Auto	—	—	—	—	—
Autoscaling	—	—	HPA	—	—	KEDA	—	—	—	—
GitOps	—	—	—	FluxCD	—	—	—	—	—	—
CVE scan	—	—	—	1 CRIT	—	—	—	—	—	—
Helm	—	—	—	—	—	—	v0.1.0	—	—	—
Policy	—	—	—	—	—	—	—	34 viol.	—	—
Chaos	—	—	—	—	—	—	—	—	pod-kill	—
mTLS	—	—	—	—	—	—	—	—	—	2/2
Issues	6	8	10	11	12	13	14	15	16	17

Troubleshooting Log — 17 incydentów

ISSUE-001 — KQL: CPUCapacityNanoCores missing

Rozwiązanie: Użyj getschema do weryfikacji kolumn

ISSUE-002 — Azure Policy: missing params

Rozwiązanie: Dodaj --params

ISSUE-003 — Terraform: RG already exists

Rozwiązanie: terraform import

ISSUE-004 — Terraform: stale plan

Rozwiązanie: Wygeneruj nowy plan po imporcie

ISSUE-005 — Container Insights after restart

Rozwiązanie: Reinstaluj az k8s-extension

ISSUE-006 — etcd Split-Brain ks3 vs k3s

Rozwiązanie: kubectl delete node + ponowna rejestracja

ISSUE-007 — Azure DevOps: TerraformInstaller

Rozwiązanie: Instalacja z Marketplace

ISSUE-008 — Pipeline: Variable Group perm

Rozwiązanie: View → Permit

ISSUE-009 — sudo ~ expansion

Rozwiązanie: Użyj pełnej ścieżki /home/buth11/

ISSUE-010 — HPA cpu: unknown

Rozwiązanie: Dodaj resources.requests

ISSUE-011 — RBAC: apiRef vs apiGroup

Rozwiązanie: Popraw na apiGroup: rbac.authorization.k8s.io

ISSUE-012 — cert-manager: duration < 1h

Rozwiązanie: Min duration 1h5m, renewBefore 6m

ISSUE-013 — KEDA: conflict with HPA

Rozwiązanie: Usun HPA przed KEDA

ISSUE-014 — Helm: SA already exists

Rozwiązanie: Usun stary SA przed instalacją

ISSUE-015 — OPA: custom template blocked

Rozwiązanie: Azure Arc zarządza Gatekeeper

ISSUE-016 — Chaos Mesh: scheduler removed

Rozwiązanie: Użyj duration zamiast scheduler

ISSUE-017 — Linkerd: no sh in container

Rozwiązanie: Użyj kontenera linkerd-proxy